FDID Package Demonstration

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Example 1: Test on Cao et al. (2022)

```r
# Load the FDID package and other necessary packages
library(fdid)
library(readstata13)

# Load the dataset (currently in long-format)
d <- as.data.frame(read.csv("mortality.csv"))

# Data Preparation
d$uniqueid <- paste(as.character(d$provid), as.character(d$countyid), sep = "-")
d$provid <- as.factor(d$provid)
d$highzupu <- ifelse(d$pczupu >= median(d$pczupu, na.rm = TRUE), 1, 0)

# Define common parameters
tr_period <- 1958:1961
ref_period <- 1954:1957
treat <- "highzupu"
Y <- "mortality"
time <- "year"
unit <- "uniqueid"
covar <- c("avggrain", "nograin", "urban", "dis_bj", "dis_pc",
           "rice", "minority", "edu", "lnpop")
```

Data Preparation

```r
prepared_data <- fdid_prepare(
  data = d,
  Y_label = Y,
  X_labels = covar,
  G_label = treat,
  unit_label = unit,
  time_label = time
)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
cat("Prepared Data Summary:\n")
```

```
## Prepared Data Summary:
```

```r
str(prepared_data)
```

```
## tibble [921 x 24] (S3: tbl_df/tbl/data.frame)
##  $ unit: chr [1:921] "13-5" "13-8" "61-11" "52-12" ...
##  $ 1954: num [1:921] 17.1 14.4 11.3 12.4 11.7 ...
##  $ 1955: num [1:921] 17.1 10.2 11.3 14.5 10.9 ...
##  $ 1956: num [1:921] 17.09 13.6 10.73 7.26 14.82 ...
##  $ 1957: num [1:921] 17.02 10.96 10.22 9.75 11.56 ...
##  $ 1958: num [1:921] 16.94 11.04 10.12 8.94 11.04 ...
##  $ 1959: num [1:921] 17.1 11.3 17.6 10.7 13 ...
##  $ 1960: num [1:921] 17 14.4 24.5 13.7 28.5 ...
##  $ 1961: num [1:921] 10.42 12.59 7.78 18.9 6.54 ...
##  $ 1962: num [1:921] 6.88 8.99 8.15 10.83 6.52 ...
##  $ 1963: num [1:921] 7.43 10.78 10.13 15.35 7.71 ...
##  $ 1964: num [1:921] 8.89 8.1 15.91 20.64 5.54 ...
##  $ 1965: num [1:921] 7.23 8.71 14.24 15.45 8.52 ...
##  $ 1966: num [1:921] 7.34 8.89 15.19 13.64 7.47 ...
##  $ x1  : num [1:921] 177 311 186 303 480 ...
##  $ x2  : num [1:921] 31.8 28.5 14 16.9 22.4 ...
##  $ x3  : num [1:921] 5 4.09 10.02 7.41 8.62 ...
##  $ x4  : num [1:921] 80 173 1008 1912 975 ...
##  $ x5  : num [1:921] 75 193 209 209 116 ...
##  $ x6  : num [1:921] 0 0 0 1 1 0 0 0 1 1 ...
##  $ x7  : num [1:921] 0 0 0 1 0 0 0 0 1 0 ...
##  $ x8  : num [1:921] 2.55 2.63 2.08 1.8 1.97 ...
##  $ x9  : num [1:921] 12.5 12.4 13.1 12.1 12.6 ...
##  $ G   : num [1:921] 1 0 0 0 0 0 1 0 0 1 ...
```

FDID Estimation

```r
# FDID estimation with DID
fdid_results_did <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "did",
  vartype = "robust",
)
```

```
## Loading required package: iterators
```

```
## Loading required package: parallel
```

```r
cat("FDID Results Summary (DID):\n")
```

```
## FDID Results Summary (DID):
```

```r
summary(fdid_results_did)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## ---------------------------------------------------
## Method:             did
## Variance Type:      robust
```

```
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error    CI_Lower  CI_Upper
## 1 0.2414733 0.1620883 -0.07663283 0.5595795
## Event Period Estimate:
##    Estimate Std.Error  CI_Lower   CI_Upper
## 1 -2.316266 0.7902333 -3.867137 -0.7653948
## Post-Event Period Estimate:
##     Estimate Std.Error  CI_Lower   CI_Upper
## 1 -0.8068464 0.2039479 -1.207104 -0.4065888
##
## Dynamic Estimates:
##         Estimate Std.Error   CI_Lower   CI_Upper
## 1954  1.04968636 0.3191425  0.4233537  1.6760191
## 1955  0.04892625 0.2764560 -0.4936320  0.5914845
## 1956 -0.13271932 0.2391295 -0.6020226  0.3365840
## 1957  0.00000000 0.0000000  0.0000000  0.0000000
## 1958 -0.35424003 0.3919047 -1.1233721  0.4148921
## 1959 -2.38137612 1.0538750 -4.4496571 -0.3130951
## 1960 -4.97749854 1.7323680 -8.3773550 -1.5776421
## 1961 -1.55194964 0.7255377 -2.9758527 -0.1280466
## 1962 -0.11016377 0.2735847 -0.6470871  0.4267596
## 1963 -0.94407885 0.2595983 -1.4535532 -0.4346045
## 1964 -1.40631166 0.2613434 -1.9192108 -0.8934125
## 1965 -0.88495265 0.2204008 -1.3174999 -0.4524054
## 1966 -0.68872498 0.2220798 -1.1245674 -0.2528825
```

```r
# FDID estimation with OLS1
fdid_results_ols1 <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ols1",
  vartype = "robust"
)
cat("FDID Results Summary (OLS1):\n")
```

```
## FDID Results Summary (OLS1):
```

```r
summary(fdid_results_ols1)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## ---------------------------------------------------
## Method:             ols1
## Variance Type:      robust
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
```

```
##    Estimate Std.Error    CI_Lower  CI_Upper
## 1 0.2500762 0.1577355 -0.05949149 0.5596439
## Event Period Estimate:
##    Estimate Std.Error CI_Lower  CI_Upper
## 1 -2.802417 0.7511273 -4.27656 -1.328273
## Post-Event Period Estimate:
##     Estimate Std.Error   CI_Lower    CI_Upper
## 1 -0.4892946 0.2024538 -0.8866253 -0.09196391
##
## Dynamic Estimates:
##        Estimate Std.Error   CI_Lower    CI_Upper
## 1954  0.9565101 0.2716033  0.4234684  1.48955177
## 1955  0.2512280 0.2663451 -0.2714940  0.77395002
## 1956 -0.2074334 0.2696620 -0.7366651  0.32179833
## 1957  0.0000000 0.0000000  0.0000000  0.00000000
## 1958 -0.8928047 0.3800214 -1.6386250 -0.14698446
## 1959 -3.5169360 1.1050783 -5.6857343 -1.34813777
## 1960 -5.6337565 1.6780712 -8.9270959 -2.34041699
## 1961 -1.1661688 0.7262691 -2.5915260  0.25918829
## 1962 -0.1948013 0.3285200 -0.8395463  0.44994366
## 1963 -0.5807112 0.2600656 -1.0911092 -0.07031324
## 1964 -0.7400733 0.2537360 -1.2380489 -0.24209761
## 1965 -0.4843303 0.2184982 -0.9131492 -0.05551143
## 1966 -0.4465568 0.2183432 -0.8750716 -0.01804197
```

```r
# FDID estimation with OLS2
fdid_results_ols2 <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ols2",
  vartype = "robust"
)
cat("FDID Results Summary (OLS2):\n")
```

```
## FDID Results Summary (OLS2):
```

```r
summary(fdid_results_ols2)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## ----------------------------------------------------
## Method:             ols2
## Variance Type:      robust
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error    CI_Lower  CI_Upper
## 1 0.2641554 0.1629352 -0.05519757 0.5835084
## Event Period Estimate:
##    Estimate Std.Error  CI_Lower  CI_Upper
## 1 -2.926352 0.7843902 -4.463756 -1.388947
## Post-Event Period Estimate:
```

```
##     Estimate Std.Error   CI_Lower    CI_Upper
## 1 -0.5085443 0.2127965 -0.9256254 -0.09146328
##
## Dynamic Estimates:
##        Estimate Std.Error   CI_Lower     CI_Upper
## 1954  0.9947531 0.2741431  0.4574326  1.532073671
## 1955  0.2735005 0.2812737 -0.2777959  0.824796835
## 1956 -0.2116320 0.2721656 -0.7450766  0.321812633
## 1957  0.0000000 0.0000000  0.0000000  0.000000000
## 1958 -0.9478755 0.3910600 -1.7143530 -0.181397970
## 1959 -3.5576406 1.1488593 -5.8094050 -1.305876335
## 1960 -5.7908962 1.7279054 -9.1775908 -2.404201679
## 1961 -1.4089942 0.7727382 -2.9235612  0.105572743
## 1962 -0.1545390 0.3628858 -0.8657953  0.556717212
## 1963 -0.6249517 0.2674773 -1.1492072 -0.100696334
## 1964 -0.8069750 0.2672727 -1.3308295 -0.283120434
## 1965 -0.5087941 0.2261796 -0.9521062 -0.065481961
## 1966 -0.4474619 0.2258399 -0.8901082 -0.004815641
```

```r
# FDID estimation with OLS2 and targetting ATT
fdid_results_ols2 <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ols2",
  vartype = "robust",
  ATT = TRUE
)
cat("FDID Results Summary (Targetting ATT with OLS2):\n")
```

```
## FDID Results Summary (Targetting ATT with OLS2):
```

```r
summary(fdid_results_ols2)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## ---------------------------------------------------
## Method:             ols2
## Variance Type:      robust
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error    CI_Lower  CI_Upper
## 1 0.2971784 0.1767948 -0.04933934 0.6436961
## Event Period Estimate:
##    Estimate Std.Error CI_Lower  CI_Upper
## 1 -3.065233 0.8197381 -4.67192 -1.458547
## Post-Event Period Estimate:
##     Estimate Std.Error   CI_Lower   CI_Upper
## 1 -0.5612293 0.2115554 -0.9758778 -0.1465808
##
## Dynamic Estimates:
##        Estimate Std.Error    CI_Lower     CI_Upper
```

```
## 1954  1.2167282 0.3274012   0.5750218  1.858434552
## 1955  0.2300345 0.2839073  -0.3264239  0.786492779
## 1956 -0.2580492 0.2945436  -0.8353546  0.319256307
## 1957  0.0000000 0.0000000   0.0000000  0.000000000
## 1958 -0.8480202 0.4315231  -1.6938054 -0.002234985
## 1959 -3.7990428 1.1893332  -6.1301360 -1.467949659
## 1960 -6.5716948 1.8184050 -10.1357686 -3.007621114
## 1961 -1.0421755 0.7894600  -2.5895170  0.505165993
## 1962 -0.3725481 0.3293819  -1.0181366  0.273040502
## 1963 -0.5883423 0.2665206  -1.1107228 -0.065961857
## 1964 -0.8479577 0.2658111  -1.3689474 -0.326968069
## 1965 -0.5035937 0.2283721  -0.9512031 -0.055984335
## 1966 -0.4937045 0.2290235  -0.9425905 -0.044818477
```

```r
# FDID with Entropy Balancing
fdid_results_ebal <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ebal",
  vartype = "robust",
  ATT = TRUE
)
cat("FDID Results Summary (Entropy Balancing):\n")
```

```
## FDID Results Summary (Entropy Balancing):
```

```r
summary(fdid_results_ebal)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## ----------------------------------------------------
## Method:             ebal
## Variance Type:      robust
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error   CI_Lower  CI_Upper
## 1 0.3870596 0.1839695 0.02601053 0.7481086
## Event Period Estimate:
##    Estimate Std.Error  CI_Lower  CI_Upper
## 1 -3.123241   0.98541 -5.057157 -1.189326
## Post-Event Period Estimate:
##     Estimate Std.Error   CI_Lower   CI_Upper
## 1 -0.5270505 0.2081339 -0.9355234 -0.1185777
##
## Dynamic Estimates:
##        Estimate Std.Error   CI_Lower     CI_Upper
## 1954  1.3229702 0.3352746  0.6649775  1.980962816
## 1955  0.4261647 0.2915152 -0.1459481  0.998277500
## 1956 -0.2008966 0.3163536 -0.8217560  0.419962830
## 1957  0.0000000 0.0000000  0.0000000  0.000000000
## 1958 -0.9299153 0.5026977 -1.9164840  0.056653504
```

```
## 1959 -4.1841573 1.3921259 -6.9162722 -1.452042340
## 1960 -6.0011348 1.9965582 -9.9194775 -2.082792024
## 1961 -1.3777586 0.8320336 -3.0106651  0.255147856
## 1962 -0.4393089 0.3463111 -1.1189614  0.240343498
## 1963 -0.5035548 0.2514204 -0.9969795 -0.010130090
## 1964 -0.7635069 0.2589691 -1.2717464 -0.255267304
## 1965 -0.4864463 0.2248689 -0.9277625 -0.045130092
## 1966 -0.4424358 0.2227823 -0.8796569 -0.005214676
```

```r
# FDID with AIPW
fdid_results_aipw <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "aipw",
  vartype = "robust"
)
cat("FDID Results Summary (AIPW):\n")
```

```
## FDID Results Summary (AIPW):
```

```r
summary(fdid_results_aipw)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## --------------------------------------------------
## Method:             aipw
## Variance Type:      robust
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error    CI_Lower  CI_Upper
## 1 0.2276032 0.1453512 -0.05728502 0.5124915
## Event Period Estimate:
##    Estimate Std.Error CI_Lower  CI_Upper
## 1 -2.523014 0.6050029 -3.70882 -1.337208
## Post-Event Period Estimate:
##     Estimate Std.Error  CI_Lower    CI_Upper
## 1 -0.4353487 0.1851395 -0.798222 -0.07247537
##
## Dynamic Estimates:
##        Estimate Std.Error   CI_Lower    CI_Upper
## 1954  0.8641155 0.2729671  0.3291000  1.39913094
## 1955  0.1476799 0.2509006 -0.3440852  0.63944500
## 1956 -0.1337065 0.2375101 -0.5992262  0.33181322
## 1957  0.0000000 0.0000000  0.0000000  0.00000000
## 1958 -0.6405121 0.3040278 -1.2364065 -0.04461764
## 1959 -3.0228034 0.8330598 -4.6556006 -1.39000619
## 1960 -4.7504429 1.4403893 -7.5736059 -1.92727994
## 1961 -1.1960753 0.6179314 -2.4072208  0.01507027
## 1962 -0.0371429 0.2750759 -0.5762917  0.50200585
## 1963 -0.4823792 0.2308156 -0.9347777 -0.02998063
## 1964 -0.7080605 0.2246025 -1.1482815 -0.26783952
```

```
## 1965 -0.4714870 0.1996086 -0.8627199 -0.08025421
## 1966 -0.4500281 0.1993354 -0.8407254 -0.05933080
```

```r
# OLS1 parallel computation with jackknife variance
fdid_results_ols1_jack_parallel <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ols1",
  vartype = "jackknife",
  parallel = TRUE,
  cores = 6,
  nsims = 100
)
cat("FDID Results Summary (OLS1 Parallel Jackknife):\n")
```

```
## FDID Results Summary (OLS1 Parallel Jackknife):
```

```r
summary(fdid_results_ols1_jack_parallel)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## ---------------------------------------------------
## Method:             ols1
## Variance Type:      jackknife
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error  CI_Lower  CI_Upper
## 1 0.2500762  0.158797 0.2388993 0.2611735
## Event Period Estimate:
##    Estimate Std.Error  CI_Lower CI_Upper
## 1 -2.802417 0.7568021 -2.847944  -2.7551
## Post-Event Period Estimate:
##     Estimate Std.Error   CI_Lower   CI_Upper
## 1 -0.4892946 0.2035742 -0.5026659 -0.4765012
##
## Dynamic Estimates:
##        Estimate Std.Error   CI_Lower   CI_Upper
## 1954  0.9565101 0.2733475  0.9378469  0.9731912
## 1955  0.2512280 0.2681531  0.2339916  0.2706229
## 1956 -0.2074334 0.2712724 -0.2213538 -0.1919167
## 1957  0.0000000 0.0000000  0.0000000  0.0000000
## 1958 -0.8928047 0.3826605 -0.9177094 -0.8693230
## 1959 -3.5169360 1.1151236 -3.5674589 -3.4585850
## 1960 -5.6337565 1.6871181 -5.7266559 -5.5349952
## 1961 -1.1661688 0.7306409 -1.2068374 -1.1105738
## 1962 -0.1948013 0.3316606 -0.2095179 -0.1773730
## 1963 -0.5807112 0.2612775 -0.5955453 -0.5651640
## 1964 -0.7400733 0.2550848 -0.7567570 -0.7221483
## 1965 -0.4843303 0.2196595 -0.4995137 -0.4700910
## 1966 -0.4465568 0.2194520 -0.4608976 -0.4324178
```

```r
# OLS2 parallel computation with jackknife variance
fdid_results_ols2_jack_parallel <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ols2",
  vartype = "jackknife",
  parallel = TRUE,
  cores = 6,
  nsims = 100
)
cat("FDID Results Summary (OLS2 Parallel Jackknife):\n")
```

```
## FDID Results Summary (OLS2 Parallel Jackknife):
```

```r
summary(fdid_results_ols2_jack_parallel)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## ---------------------------------------------------
## Method:            ols2
## Variance Type:     jackknife
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error  CI_Lower  CI_Upper
## 1 0.2641554 0.1623717 0.2528503 0.2750552
## Event Period Estimate:
##    Estimate Std.Error  CI_Lower  CI_Upper
## 1 -2.926352 0.7907025 -2.973656 -2.877949
## Post-Event Period Estimate:
##     Estimate Std.Error   CI_Lower   CI_Upper
## 1 -0.5085443 0.2116717 -0.5227136 -0.4955786
##
## Dynamic Estimates:
##        Estimate Std.Error   CI_Lower   CI_Upper
## 1954  0.9947531 0.2783092  0.9764106  1.0123299
## 1955  0.2735005 0.2793394  0.2555286  0.2942501
## 1956 -0.2116320 0.2705398 -0.2254496 -0.1953014
## 1957  0.0000000 0.0000000  0.0000000  0.0000000
## 1958 -0.9478755 0.3913163 -0.9726478 -0.9226249
## 1959 -3.5576406 1.1569648 -3.6097204 -3.5012825
## 1960 -5.7908962 1.7352405 -5.8851542 -5.6882735
## 1961 -1.4089942 0.7701129 -1.4501234 -1.3527122
## 1962 -0.1545390 0.3624708 -0.1697447 -0.1369617
## 1963 -0.6249517 0.2656772 -0.6398882 -0.6085637
## 1964 -0.8069750 0.2672546 -0.8251318 -0.7895496
## 1965 -0.5087941 0.2249914 -0.5247396 -0.4945724
## 1966 -0.4474619 0.2243164 -0.4619532 -0.4322872
```

```r
# OLS1 parallel computation with bootstrap variance
fdid_results_ols1_boot_parallel <- fdid(
  s = prepared_data,
```

```r
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ols1",
  vartype = "bootstrap",
  parallel = TRUE,
  cores = 6,
  nsims = 100
)
cat("FDID Results Summary (OLS1 Parallel Bootstrap):\n")
```

```
## FDID Results Summary (OLS1 Parallel Bootstrap):
```

```r
summary(fdid_results_ols1_boot_parallel)
```

```
## Factorial Difference-in-Differences (FDID) Summary
## ----------------------------------------------------
## Method:             ols1
## Variance Type:      bootstrap
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error     CI_Lower CI_Upper
## 1 0.2500762 0.1507369 -0.002307484 0.575565
## Event Period Estimate:
##    Estimate Std.Error  CI_Lower  CI_Upper
## 1 -2.802417 0.8958385 -4.524519 -1.089433
## Post-Event Period Estimate:
##     Estimate Std.Error   CI_Lower   CI_Upper
## 1 -0.4892946 0.1797719 -0.8064631 -0.1469534
##
## Dynamic Estimates:
##        Estimate Std.Error   CI_Lower    CI_Upper
## 1954  0.9565101 0.2965365  0.3064880  1.41922110
## 1955  0.2512280 0.2897723 -0.4264906  0.82619016
## 1956 -0.2074334 0.2521967 -0.6733977  0.27030428
## 1957  0.0000000 0.0000000  0.0000000  0.00000000
## 1958 -0.8928047 0.3620174 -1.5496092 -0.22177543
## 1959 -3.5169360 1.0893182 -5.7982042 -1.32180359
## 1960 -5.6337565 1.7838636 -7.9341330 -1.73612108
## 1961 -1.1661688 0.7020276 -2.4414053  0.26882224
## 1962 -0.1948013 0.3216241 -0.7489123  0.35012733
## 1963 -0.5807112 0.2601048 -1.0404533 -0.09707645
## 1964 -0.7400733 0.2375282 -1.2196811 -0.36632766
## 1965 -0.4843303 0.2043154 -0.8831788 -0.09638072
## 1966 -0.4465568 0.2130083 -0.7554975 -0.02153940
```

```r
# OLS2 parallel computation with bootstrap variance
fdid_results_ols2_boot_parallel <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ols2",
```

```r
  vartype = "bootstrap",
  parallel = TRUE,
  cores = 6,
  nsims = 100
)
cat("FDID Results Summary (OLS2 Parallel Bootstrap):\n")
```

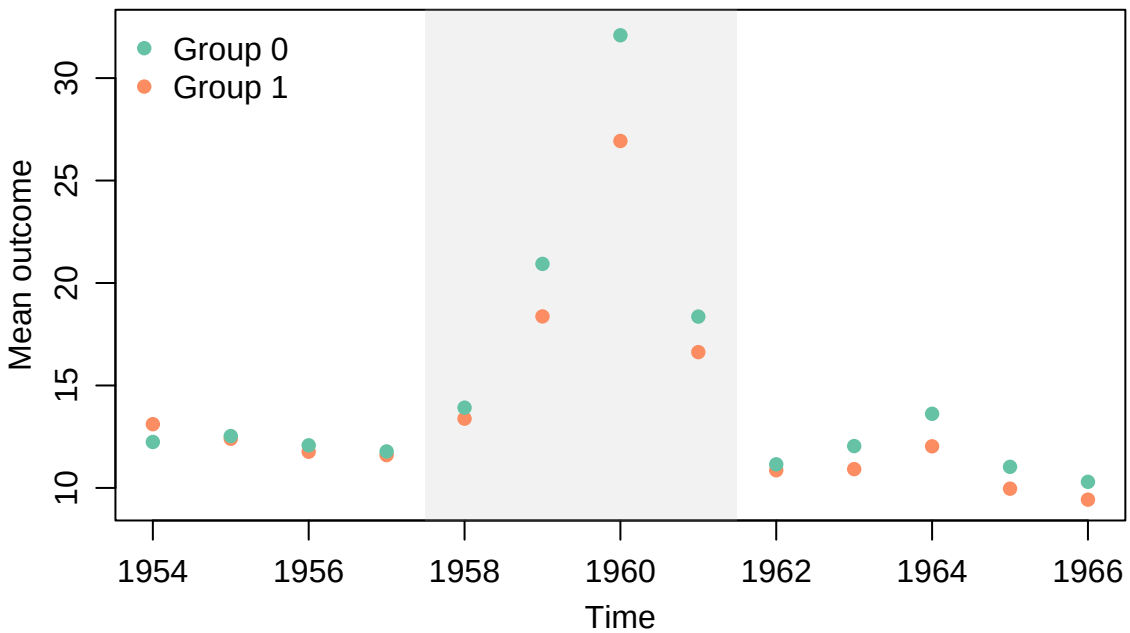
```
## FDID Results Summary (OLS2 Parallel Bootstrap):
```

```r
summary(fdid_results_ols2_boot_parallel)
```

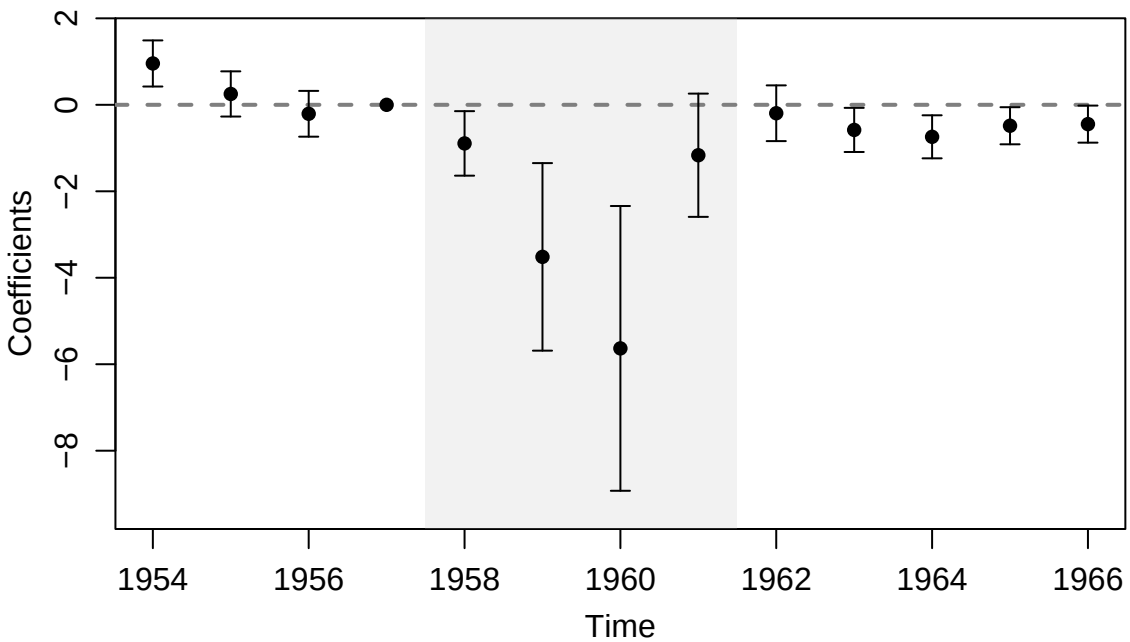
```
## Factorial Difference-in-Differences (FDID) Summary
## ---------------------------------------------------
## Method:             ols2
## Variance Type:      bootstrap
## Reference Period:  1957
##
## Pre-Event Period:  1954, 1955, 1956, 1957
## Event Period:  1958, 1959, 1960, 1961
## Post-Event Period:  1962, 1963, 1964, 1965, 1966
## Pre-Event Period Estimate:
##    Estimate Std.Error   CI_Lower  CI_Upper
## 1 0.2641554 0.1427529 0.04853943 0.5490985
## Event Period Estimate:
##    Estimate Std.Error  CI_Lower  CI_Upper
## 1 -2.926352 0.7727365 -4.323477 -1.658683
## Post-Event Period Estimate:
##     Estimate Std.Error   CI_Lower   CI_Upper
## 1 -0.5085443 0.2288785 -0.8913108 -0.0550937
##
## Dynamic Estimates:
##        Estimate Std.Error   CI_Lower     CI_Upper
## 1954  0.9947531 0.2753283  0.4388098  1.520180913
## 1955  0.2735005 0.2454181 -0.1746167  0.782334927
## 1956 -0.2116320 0.2831752 -0.6996595  0.372890407
## 1957  0.0000000 0.0000000  0.0000000  0.000000000
## 1958 -0.9478755 0.4042206 -1.9152558 -0.348593387
## 1959 -3.5576406 1.1552486 -5.6998870 -1.332391305
## 1960 -5.7908962 1.6198743 -9.1523938 -3.347714366
## 1961 -1.4089942 0.8347850 -3.1073571  0.458673371
## 1962 -0.1545390 0.3617401 -0.9160216  0.340198099
## 1963 -0.6249517 0.2542774 -1.0643628 -0.147225943
## 1964 -0.8069750 0.2551476 -1.3570779 -0.425261901
## 1965 -0.5087941 0.1944312 -0.8802993 -0.164995199
## 1966 -0.4474619 0.2315935 -0.8440449  0.009299047
```

Visualization

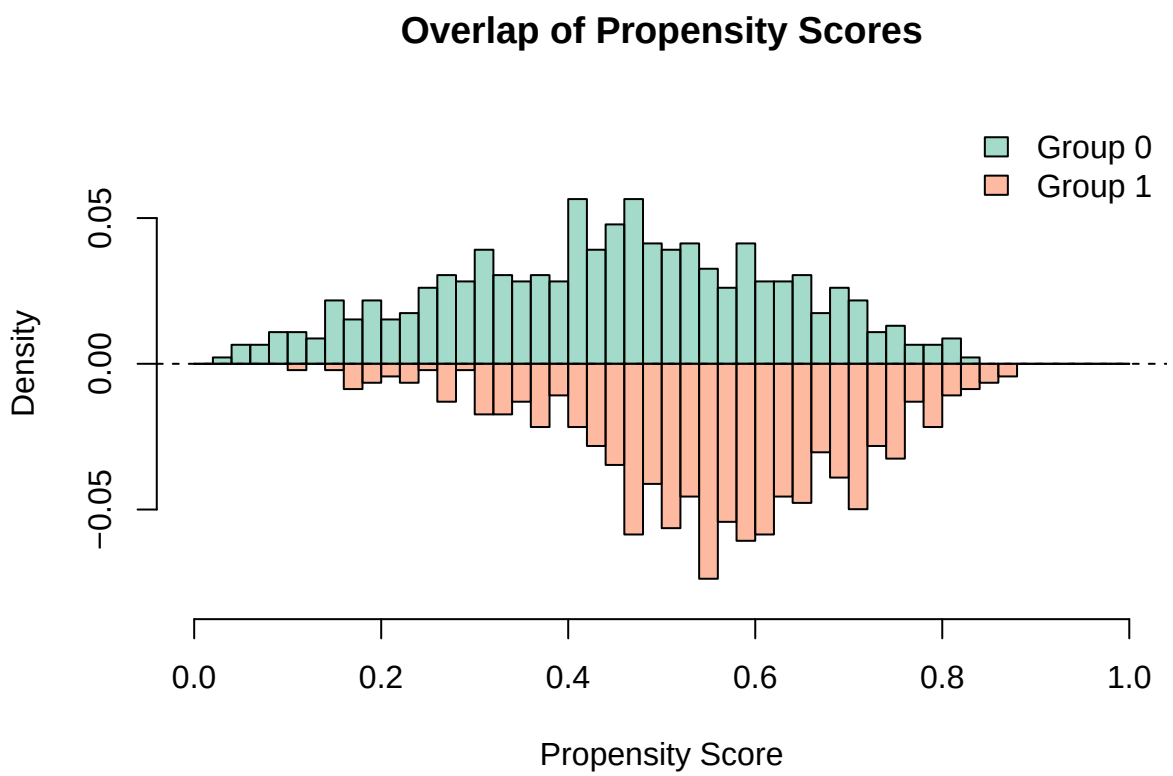
```r
# Raw means plot
plot(fdid_results_ols1, type = "raw")
```



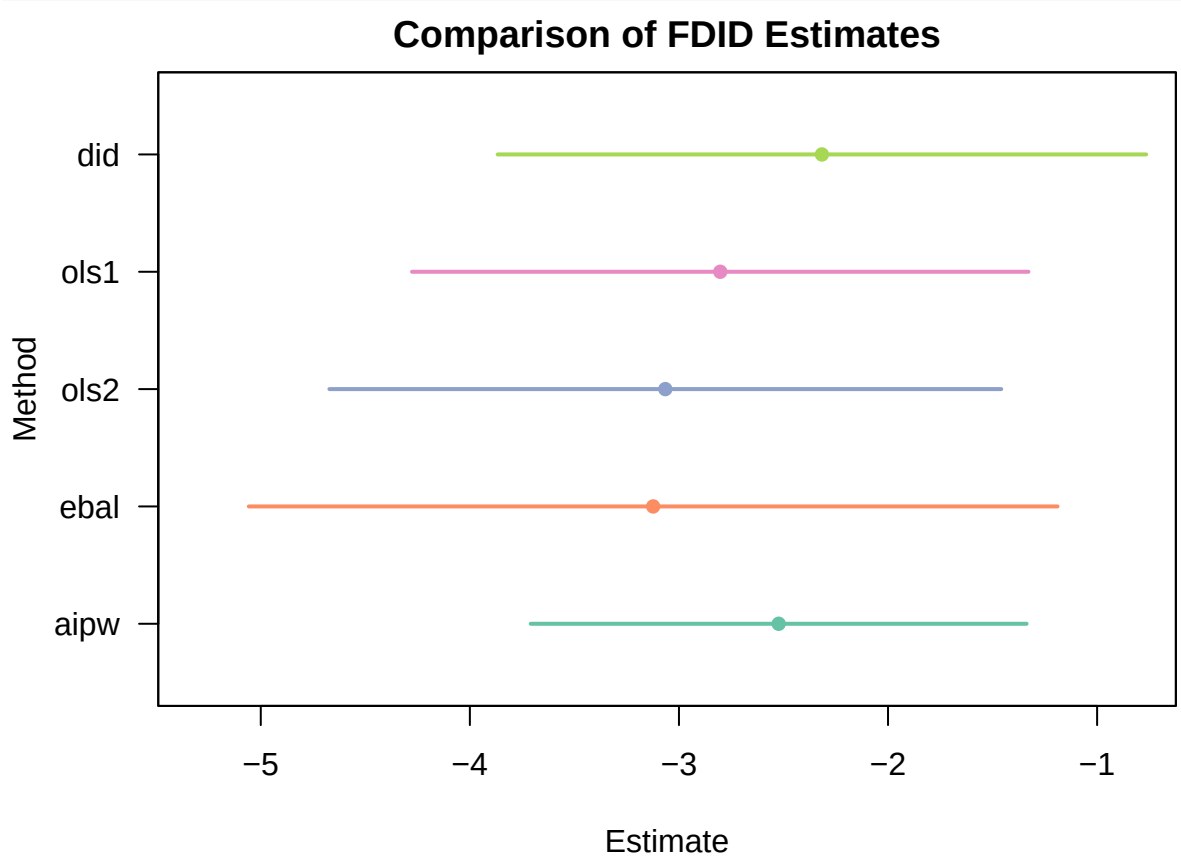

```r
# Dynamic effects plot
plot(fdid_results_ols1, type = "dynamic")
```




```r
# Propensity score overlap
plot(fdid_results_aipw, type = "overlap")
```




```r
# Comparison of estimates
plot(fdid_list(fdid_results_did, fdid_results_ols1, fdid_results_ols2, fdid_results_ebal, fdid_results_a
```

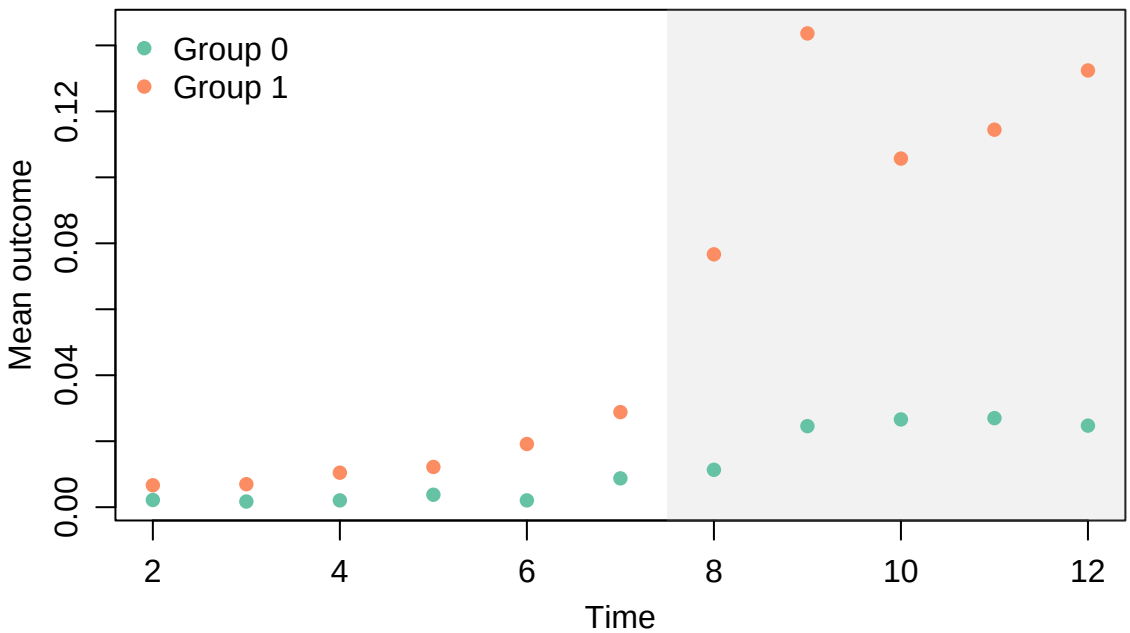


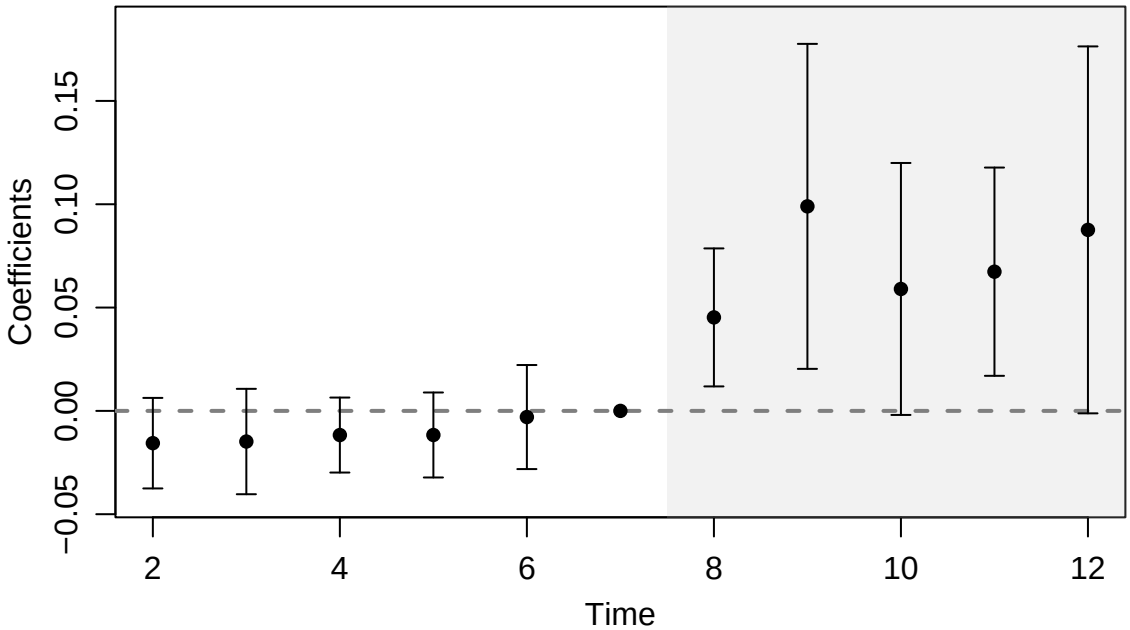
Example 2: Replication of Chen et al. (2024)

```r
# Load the dataset (currently in long-format)
d <- read.dta13("Table1.dta")

# Data Preparation
wubaos <- names(tapply(d$stronghold_reform, d$prefecture_id, mean)
                [tapply(d$stronghold_reform, d$prefecture_id, mean)>0])
d$wubao <- ifelse(d$prefecture_id %in% wubaos, 1, 0)

d$uniqueid <- paste(as.character(d$prov_id), as.character(d$prefecture_id), sep = "-")
d$prefecture_id = factor(d$prefecture_id)

# Define common parameters
tr_period <- 8:12
ref_period = 2:7
treat <- "wubao"
Y <- "arist_recruit"
time <- "emperor_id"
unit <- "uniqueid"
cluster <- "prefecture_id"
```

Plots look the same

```r
prepared_data <- fdid_prepare(
  data = d,
  Y_label = Y,
  X_labels = NULL,
  G_label = treat,
  unit_label = unit,
  time_label = time,
  cluster_label = cluster
)
fdid_results_ols1 <- fdid(
  s = prepared_data,
  tr_period = tr_period,
  ref_period = ref_period,
  method = "ols1",
  vartype = "robust"
)

plot(fdid_results_ols1)
```




```
plot(fdid_results_ols1, type="dynamic")
```